

Experiment - 7

Software Design Documentation

for

Exam Registration System

Introduction

The purpose of this document is to provide a comprehensive overview of the software design for the Exam Registration System. This document outlines the system architecture, design considerations, functional and non-functional specifications, detailed component designs, implementation strategy, testing approach, and maintenance plan. The Exam Registration System aims to streamline the process of exam enrolment for students and ensure efficient administrative control for academic institutions.

System Overview

The Exam Registration System is a web-based application designed to simplify and automate the process of exam enrolment for students and administrators. It enables students to view available exams, register for them, and receive confirmation, while providing administrators with tools to manage exam schedules, track registrations and generate reports. The system will be developed using HTML, C5Sr and JavaScript for the frontend, with a Nodejs backend to handle server-side logic and database interactions.

Design Considerations

- **User-Friendly Interface:** The system will feature a clean and intuitive interface to ensure ease to use for both students and administrators, reducing the learning curve and enhancing user experience.
- **Scalability:** The design will support scalability to accommodate a growing number of students, courses, and earns without performance degradation.
- **Security:** Robust security measures will be implemented to protect sensitive user data including secure authentication, authorization, and data encryption.
- **Cross-Browser and Cross-Device Compatibility:** The application will be compatible with all major web browsers and responsive across various devices including desktops, tablets, and smartphones.
- **Integration Capabilities:** The system will be designed with provisions For integrating with external systems or APIs, such as student information systems or email/SMS notification services.

Design Specifications

- The system shall allow students to register for available exams through a user-friendly interface
- Administrators should be able to create, edit, and delete exam schedules.
- The system Should provide real-time updates on registration status and seat availability.
- RLI User data, including personal and academic information, shall be securely stored and encrypted
- The system must be compatible with the latest versions of popular browsers (e.g., Chrome, Firefox Edge).
- Integration with coil/SMS API for sending registration confirmations and reminders.

Detailed Design

The detailed design of the Exam Registration System encompasses the system architecture, database schema, and user interface layouts. The architecture defines the interaction between various system components, including the frontend backend, and database. The database schema outlines the structure for storing student records, exam details, and registration data, The user interface design focuses on creating a seamless experience for both students and administrators. These components are elaborated in the accompanying design documents

Implementation Plan

Task	Assigned To	Start Date	End Date	Status
Frontend Development	Samyak	03/01/2025	03/15/2025	In Progress
Backend Development	Rohit	03/16/2025	04/10/2025	Not Started
Database Setup	Sahil	02/25/2025	02/28/2025	In Progress

Testing Plan

Unit Testing: Each individual component of the system (e.g., exam registration form, login module, admin panel) will undergo unit testing to verify that it functions as intended in isolation.

- Integration Testing: After unit testing, integration testing will be conducted to ensure that different modules such as student registration, exam scheduling, and notification system — work together seamlessly.
- User Acceptance Testing (IJAT): A selected group of students and administrators will test the system in a real-world scenario to validate that it meets user needs and expectations.
- Performance Testing: The system will be tested under various loads to assess its scalability and responsiveness, ensuring it can handle high numbers of concurrent users during peak exam registration periods.

Maintenance Plan

The maintenance plan for the Exam Registration System includes regular updates to ensure continued system performance, security, and relevance, this involves periodic bug fixes, performance optimizations, and feature enhancements based on User feedback and evolving academic requirements. The plan also defines a clear process for handling support requests, troubleshooting TISSUES, and maintaining high system availability, especially during critical exam registration periods. Scheduled maintenance windows will be communicated in advance to minimize disruption to users.

Conclusion

This software design documentation provides a comprehensive overview to the Exam Registration System design. It includes the system overview, design considerations, specifications, detailed design implementation plan, testing plan, and maintenance plane Any future revisions or updates to this document will be recorded in the revision history section below.

Revisions:

Version	Date	Author	Description
1.0	04/13/2025	Sahil	Initial version
1.1	04/14/2025	Samyak	Updated implementation plan
1.2	04/15/2025	Rohit	Updated maintenance plan